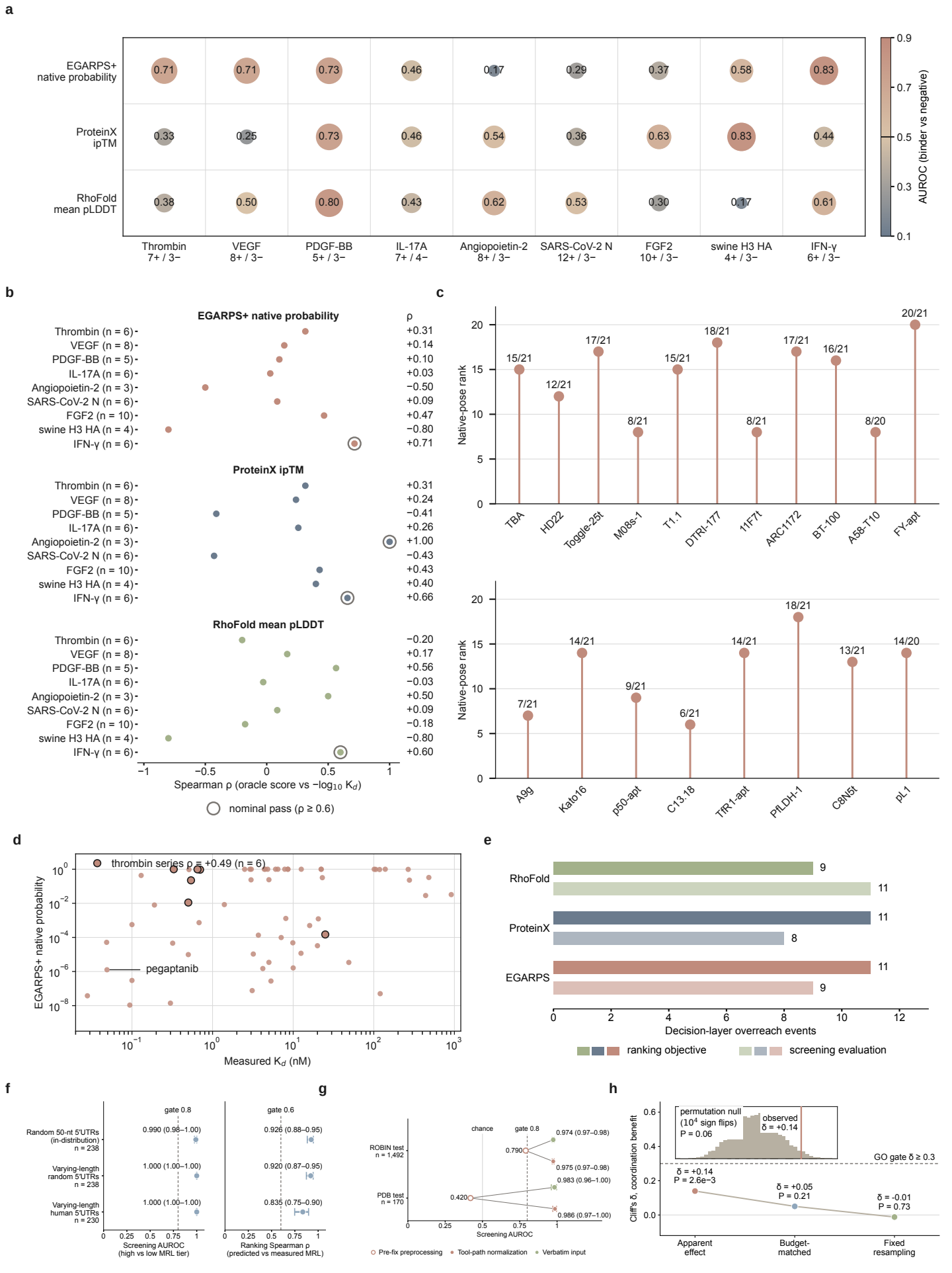

a
EGARPS+ native probability
ProteinX ipTM
RhoFold mean pLDDT
0.71 0.71 0.73 0.46 0.17 0.29 0.37 0.58 0.83
0.33 0.25 0.73 0.46 0.54 0.36 0.63 0.83 0.44
0.38 0.50 0.80 0.43 0.62 0.53 0.30 0.17 0.61
Thrombin 7+ / 3−
VEGF 8+ / 3−
PDGF-BB 5+ / 3−
IL-17A 7+ / 4−
Angiopoietin-2 8+ / 3−
SARS-CoV-2 N 12+ / 3−
FGF2 10+ / 3−
swine H3 HA 4+ / 3−
IFN-γ 6+ / 3−
AUROC (binder vs negative)
0.9
0.7
0.5
0.3
0.1
b
EGARPS+ native probability
ρ
Thrombin (n = 6) +0.31
VEGF (n = 8) +0.14
PDGF-BB (n = 5) +0.10
IL-17A (n = 6) +0.03
Angiopoietin-2 (n = 3) −0.50
SARS-CoV-2 N (n = 6) +0.09
FGF2 (n = 10) +0.47
swine H3 HA (n = 4) −0.80
IFN-γ (n = 6) +0.71
ProteinX ipTM
Thrombin (n = 6) +0.31
VEGF (n = 8) +0.24
PDGF-BB (n = 5) −0.41
IL-17A (n = 6) +0.26
Angiopoietin-2 (n = 3) +1.00
SARS-CoV-2 N (n = 6) −0.43
FGF2 (n = 10) +0.43
swine H3 HA (n = 4) +0.40
IFN-γ (n = 6) +0.66
RhoFold mean pLDDT
Thrombin (n = 6) −0.20
VEGF (n = 8) +0.17
PDGF-BB (n = 5) +0.56
IL-17A (n = 6) −0.03
Angiopoietin-2 (n = 3) +0.50
SARS-CoV-2 N (n = 6) +0.09
FGF2 (n = 10) −0.18
swine H3 HA (n = 4) −0.80
IFN-γ (n = 6) +0.60
−1 −0.5 0 0.5 1
Spearman ρ (oracle score vs −log10 Kd)
nominal pass (ρ ≥ 0.6)
c
Native-pose rank
15/21 12/21 17/21 8/21 15/21 18/21 8/21 17/21 16/21 8/20 20/21
TBA HD22 Toggle-25t M08s-1 T1.1 DTRI-177 11F7t ARC1172 BT-100 A58-T10 FY-apt
7/21 14/21 9/21 6/21 14/21 18/21 13/21 14/20
A9g Kato16 p50-apt C13.18 TfR1-apt PfLDH-1 C8N5t pL1
d
EGARPS+ native probability
thrombin series ρ = +0.49 (n = 6)
pegaptanib
Measured Kd (nM)
e
RhoFold 9 11
ProteinX 11 8
EGARPS 11 9
Decision-layer overreach events
ranking objective
screening evaluation
f
gate 0.8
gate 0.6
Random 50-nt 5′UTRs (in-distribution) n = 238
Varying-length random 5′UTRs n = 238
Varying-length human 5′UTRs n = 230
0.990 (0.96–1.00)
1.000 (1.00–1.00)
1.000 (1.00–1.00)
0.926 (0.88–0.95)
0.920 (0.87–0.95)
0.835 (0.75–0.90)
Screening AUROC (high vs low MRL tier)
Ranking Spearman ρ (predicted vs measured MRL)
g
chance
gate 0.8
ROBIN test n = 1,492
PDB test n = 170
0.790
0.420
0.974 (0.97–0.98)
0.975 (0.97–0.98)
0.983 (0.96–1.00)
0.986 (0.97–1.00)
Screening AUROC
Pre-fix preprocessing
Tool-path normalization
Verbatim input
h
permutation null (10^4 sign flips) P = 0.06
observed δ = +0.14
GO gate δ ≥ 0.3
δ = +0.14 P = 2.6e−3
δ = +0.05 P = 0.21
δ = -0.01 P = 0.73
Cliff's δ, coordination benefit
Apparent effect
Budget-matched
Fixed resampling